Problem 1 – Siege

1 The problem

Throughout, a *monoid* is an integral, saturated, sharp, commutative, unital semigroup; monoids are written additively. A monoid M induces a translation invariant partial order on its associated group M^{gp} such that $x \geq 0$ if and only if $x \in M$. A *valuative monoid* is a monoid in which the associated partial order is a total order. If M is a monoid, $x \in M^{\mathrm{gp}}$ and $y \in M$, we write $x \prec y$ to mean that there are integers m and n such that $my \leq x \leq ny$. A monoid homomorphism $f\colon M \to N$ is called *infinitesimal* if the following condition holds:

- if there are $x \in M^{\mathrm{gp}}$ and $y \in M$ such that $f(x) \in N$, $x \prec y$ and $f(y) \prec f(x)$, then $x \in M$.

If $P \xrightarrow{f} Q \xrightarrow{g} N$ are monoid homomorphisms, we say that f is *birhizomic at* g if it is of finite presentation in the category of (integral, saturated, sharp) monoids and, for every factorization of gf through monoid homomorphisms $P \xrightarrow{h} M \xrightarrow{p} N$ with p infinitesimal, there is a unique monoid homomorphism $u\colon Q \to M$ with $pu = g$ and $uf = h$. A family of morphisms $P \to Q_i$ is called a *birhizomic cover* if, for every valuative monoid V and every homomorphism $v\colon P \to V$, there is a factorization of v through some $Q_i \to V$ such that $P \to Q_i$ is birhizomic at $Q_i \to V$.

Let P be a monoid and let A_P be the category of monoid homomorphisms $P \to Q$. The birhizomic covers form a Grothendieck topology on A_P^{op}. For any $x \in P$ there is an initial monoid homomorphism $P \to P[-x]^\sharp$ that sends x to 0. Let B_P be the full subcategory of A_P consisting of the homomorphisms $P \to P[-x]^\sharp$, $x \in P$. Give A_P^{op} the birhizomic topology and B_P^{op} the trivial topology, in which all presheaves are sheaves. The inclusion $B_P^{\mathrm{op}} \subset A_P^{\mathrm{op}}$ is left exact; by left Kan extension and sheafification it extends to a functor $\pi^{-1}\colon \mathrm{Sh}(B_P) \to \mathrm{Sh}(A_P)$ on the categories of sheaves. Let π_* be the right adjoint of π^{-1}.

Question. *Is it the case that, for all presheaves of abelian groups F on B_P^{op}, the morphism $F \to \mathrm{R}\pi_*\pi^{-1}F$ is a quasi-isomorphism?*

2 The result

The answer is **yes**, for every monoid P.

Theorem 2.1 (Main Theorem)**.** *Let P be a monoid (integral, saturated, sharp, commutative; P is* not *assumed to be finitely generated). For every presheaf of abelian groups F on B_P^{op}:*

(a) the unit $F \to \pi_*\pi^{-1}F$ is an isomorphism;

(b) $\mathrm{R}^p\pi_*\pi^{-1}F = 0$ for all $p > 0$.

Consequently $F \to \mathrm{R}\pi_*\pi^{-1}F$ is a quasi-isomorphism.

Convention 2.2 (Reading of the problem)**.** We make the following natural readings explicit.

(i) The problem defines birhizomic covers only with base P. For an object $P \to Q$ of A_P, the covering families of Q are the birhizomic covers $\{Q \to Q_i\}$ with base Q (a map under P out of Q is just a map out of Q), and the topology on A_P^{op} is the one these families generate. Lemma 5.2 checks that they form a pretopology. The argument below only uses property ($\star$) of Section 5.

(ii) "Of finite presentation in the category of monoids" is read as compactness in the undercategory; see Definition 4.4.

(iii) $\pi^{-1}F = (\operatorname{Lan} F)^{\#}$, where Lan is the left Kan extension of presheaves and $\#$ is sheafification, as described in the problem.

(iv) *Size.* Fix universes $\mathbb{U} \in \mathbb{V}$ with $P \in \mathbb{U}$; A_P consists of the maps $P \to Q$ with $Q \in \mathbb{U}$, and presheaves take values in $\mathbb{V}$-small abelian groups, so the usual theory (plus construction, enough injectives, derived functors) applies to the site A_P^{op} [1, Exp. I]. Nothing below depends on this choice.

Remark 2.3. The proof never uses the infinitesimality condition in the definition of "birhizomic". It uses only two properties of a birhizomic cover $\{Q \to Q_i\}$: every valuation of Q (a homomorphism to a valuative monoid) factors through some member $Q \to Q_i$ at which that member is birhizomic, and such members are of finite presentation.

3 Overview of the proof

By the universal property of $P[-x]^\sharp$, the category B_P is a preorder, equivalent to the directed poset $\Pr(P)$ of principal faces $\operatorname{face}(x) = \{y \in P : y \le nx \text{ for some } n\}$ of P. For an object $f\colon P \to Q$ of A_P, the value $(\operatorname{Lan} F)(Q)$ is the filtered colimit of F over the principal faces contained in the *kernel face* $\Phi_Q = f^{-1}(0)$. Two algebraic results drive the proof.

Theorem 6.1: if Q is of finite presentation under P, then Φ_Q is principal, so $(\operatorname{Lan} F)(Q) = F(\Phi_Q)$ is a single value of F. The proof writes Q as a retract of a finitely presented reflection, describes Φ_Q through a finitely generated monoid of relations (Gordan's lemma), and uses a rank argument in a finitely generated free abelian group.

Theorem 7.2 (nerve theorem): for a family of finitely presented maps $Q \to Q_i$ through which every valuation of Q factors, and for a subset $\tau \supset \Phi_Q$, the simplicial complex of those finite σ whose pushout Q_σ has kernel face inside τ is acyclic. We translate this into convex geometry over $\mathbb{Q}$. Valuations of Q correspond to *total cones* W in $Q^{\mathrm{gp}} \otimes \mathbb{Q}$. By Fourier–Motzkin elimination, the total cones W whose valuation v_W extends to Q_i are exactly those containing a certain finite set X_i. The kernel faces of the pushouts are read off from the lineality spaces of the cones generated by the X_i. Compactness of the space of total cones yields a finite subcover. Finally an induction on the number of undecided sign conditions, using a double complex, proves acyclicity of an abstract cochain complex built from cones (Theorem 7.7).

The sheaf-theoretic part (Section 8) then runs as follows. Presheaves on B_P have injective resolutions by products of simple injectives $I_{\tau,A}$. By Theorem 7.2, the left Kan extensions of these have exact Čech complexes on all covers of finitely presented objects by finitely presented maps. From this we deduce that $\operatorname{Lan} F$ already satisfies the sheaf condition on finitely presented objects, so sheafification does not change its values there; this gives the unit isomorphism. A version of Cartan's criterion, proved here, shows that the sheaves $\pi^{-1}I^k$ have no higher cohomology on finitely presented objects. Since every object of B_P is finitely presented, the higher derived functors $\mathrm{R}^p\pi_*\pi^{-1}F$ vanish.

What is known and what is argued here. We use, as known results: Zorn's lemma, Tychonoff's theorem, Gordan's lemma [3, 2], Fourier–Motzkin elimination [7] (the version over ordered $\mathbb{Q}$-vector spaces is argued below), the universal coefficient theorem [5], the acyclic assembly lemma for double complexes [8], Kan extensions [6], the passage from a pretopology to a topology, the plus construction and enough injectives for sheaves [1, 4]. Theorems 6.1, 7.2, 7.7, the version of Cartan's criterion (Proposition 8.9) and the remaining steps are proved in full below.

4 Monoids

4.1 Notation

Let $\mathcal{C}$ be the category of monoids (integral, saturated, sharp, commutative). For a monoid B, $B/\mathcal{C}$ is the category of maps $B \to M$ in $\mathcal{C}$, with maps under B as morphisms; its Hom-sets are written $\mathrm{Hom}_B(M,N)$. For $f\colon P \to Q$ write $\Phi_Q := f^{-1}(0)$. On P^{gp}, $y \le z$ means $z - y \in P$. For $x \in P$ put $\mathrm{face}(x) := \{y \in P : y \le nx \text{ for some } n \ge 1\}$. A *face* of P is a submonoid F with $a + b \in F \Rightarrow a, b \in F$; it is *principal* if $F = \mathrm{face}(x)$ for some x. Write $\mathrm{Pr}(P)$ for the set of principal faces.

Lemma 4.1 (Torsion-freeness)**.** *For $M \in \mathcal{C}$, M^{gp} is torsion-free. Consequently $nx = ny \Rightarrow x = y$ in M^{gp} for $n \ge 1$, and $M \subset M^{\mathrm{gp}} \subset M^{\mathrm{gp}} \otimes \mathbb{Q}$.*

Proof. If $nx = 0$ then $nx \in M$, so $x \in M$ by saturation; likewise $-x \in M$, so $x = 0$ by sharpness. □

Lemma 4.2 (Filtered colimits)**.** *A filtered colimit of objects of $\mathcal{C}$, computed in commutative monoids, lies in $\mathcal{C}$. Hence filtered colimits in $\mathcal{C}$ and in $B/\mathcal{C}$ are computed on underlying sets.*

Proof. Each axiom is witnessed at a finite stage. *Integrality:* if $a + c = b + c$ in the colimit, this already holds at some stage, so $a = b$ there. *Saturation:* the group of the colimit is the colimit of the groups, and an element x of it is a difference $x = a - b$ of elements a, b of the colimit monoid, which come from some stage k. If $nx = m$ with m in the colimit monoid, then $na = nb + m$ holds in the monoid at some stage $k' \ge k$ where m is also represented. At that stage $a - b$ lies in the group of the stage monoid and $n(a - b)$ lies in the stage monoid, so $a - b$ lies in the stage monoid by saturation; hence x lies in the colimit monoid. *Sharpness:* $x + y = 0$ already holds at some stage. □

Construction 4.3 (The reflection; pushouts)**.** Let N be a commutative monoid and $E \subset N \times N$ a set of relations. Let G be the group completion of the quotient monoid N/E, M_0 the image of N/E in G, $M_1 = \{g \in G : ng \in M_0 \text{ for some } n \ge 1\}$, and $T := M_1/M_1^*$. We write $T = (N/E)^{\mathrm{int,sat,sharp}}$.

- $T \in \mathcal{C}$. It is integral as a submonoid of the group G/M_1^*, and it is sharp. It is saturated: if $ng \in T$ for $g \in G/M_1^*$, then a lift $\tilde{g}$ has $n\tilde{g} \in M_1$, so $\tilde{g} \in M_1$.
- Every map $N \to X$ with $X \in \mathcal{C}$ that respects E factors uniquely through T: it extends to $G \to X^{\mathrm{gp}}$; it sends M_1 into X because X is saturated (and X^{gp} is torsion-free); and it kills M_1^* because X is sharp.

Pushouts in $\mathcal{C}$ are formed in this way from pushouts of commutative monoids. Fibre products in A_P^{op} are pushouts under P.

Definition 4.4. A map $B \to M$ in $\mathcal{C}$ is *of finite presentation* (*f.p.*) if $\mathrm{Hom}_B(M, -)$ commutes with filtered colimits in $B/\mathcal{C}$. An object $P \to Q$ of A_P is called f.p. if this map is f.p.

Lemma 4.5 (The objects b_x)**.** *Let $x \in P$.*

(a) $P[-x] := \{p - nx : p \in P,\ n \geq 0\} \subset P^{\mathrm{gp}}$ *is integral and saturated, and* $P[-x]^\sharp := P[-x]/P[-x]^*$ *lies in $\mathcal{C}$. Write b_x for the object $P \to P[-x]^\sharp$ of A_P.*

(b) For $f\colon P \to Q$ in A_P, a map $P[-x]^\sharp \to Q$ under P exists if and only if $f(x) = 0$, and it is then unique.

(c) $\Phi_{b_x} = \mathrm{face}(x)$.

(d) For every $f\colon P \to Q$, Φ_Q is a face; a map $Q \to Q'$ under P gives $\Phi_Q \subset \Phi_{Q'}$; and $x \in \Phi_Q$ iff $\mathrm{face}(x) \subset \Phi_Q$.

(e) $\mathrm{Hom}_{A_P}(b_x, b_y)$ *is a point if* $\mathrm{face}(x) \subset \mathrm{face}(y)$ *and empty otherwise. So B_P is a preorder, equivalent to the poset* $\mathrm{Pr}(P)$*. This poset is directed, since* $\mathrm{face}(x), \mathrm{face}(y) \subset \mathrm{face}(x+y)$*, and has least element* $\mathrm{face}(0) = \{0\}$*, with $b_0 = P$.*

For $\tau \in \mathrm{Pr}(P)$ we write b_τ for any b_x with $\mathrm{face}(x) = \tau$*, and $F(\tau) := F(b_\tau)$ for a presheaf F on B_P^{op}.*

Proof. (a) Integrality is clear. If $mg \in P[-x]$ for $g \in P^{\mathrm{gp}}$, then $m(g + kx) \in P$ for some k, so $g + kx \in P$ by saturation of P, i.e. $g \in P[-x]$. A quotient of an integral saturated monoid by its units is in $\mathcal{C}$. (b) If $f(x) = 0$, f extends uniquely to $P[-x]$, which is generated by P and $-x$, and the extension kills units because Q is sharp. Conversely x is a unit in $P[-x]$, so it maps to 0 in $P[-x]^\sharp$. (c) $y \in P$ maps to 0 iff y is a unit of $P[-x]$, iff $-y = p - nx$ for some $p \in P$, $n \geq 0$, iff $y + p = nx$, iff $y \leq nx$. (d) $f(a) + f(b) = 0$ forces $f(a) = f(b) = 0$ in the sharp monoid Q. The other two statements are clear (for the last, $\mathrm{face}(x) \ni y$ with $y \leq nx$ and $nx \in \Phi_Q$ give $y \in \Phi_Q$ since Φ_Q is a face). (e) By (b), a map $b_x \to b_y$ exists iff $x \in \Phi_{b_y} = \mathrm{face}(y)$, iff $\mathrm{face}(x) \subset \mathrm{face}(y)$, and it is unique. □

Lemma 4.6. *The following hold.*

(a) Composites of f.p. maps are f.p.

(b) If $Q \to Q'$ is f.p. and $Q \to R$ is arbitrary, then $R \to Q' \sqcup_Q R$ is f.p.

(c) Each $P \to P[-x]^\sharp$ is f.p.

Hence every finite iterated pushout over Q of f.p. maps $Q \to Q_i$, with Q f.p. under P, is f.p. under P.

Proof. (b) $\mathrm{Hom}_R(Q' \sqcup_Q R, X) = \mathrm{Hom}_Q(Q', X)$, and a filtered colimit under R is also one under Q.

(a) Let $P \to Q \to Q'$ be f.p. and $X = \mathrm{colim}_k X_k$ a filtered colimit under P. A map $Q' \to X$ under P is a pair (a, c) of a map $a\colon Q \to X$ under P and a map $c\colon Q' \to X$ under Q, where X is viewed under Q via a. *Surjectivity of* $\mathrm{colim}_k \mathrm{Hom}_P(Q', X_k) \to \mathrm{Hom}_P(Q', X)$*:* a factors through some X_k (Q is f.p. under P); the $X_{k'}$ with $k' \geq k$, viewed under Q via a, form a filtered system with colimit X under Q, so c factors through some $X_{k'}$ (Q' is f.p. under Q). *Injectivity:* let $c_1, c_2\colon Q' \to X_k$ be maps under P with the same image in $\mathrm{Hom}_P(Q', X)$. Their restrictions to Q have the same image in $\mathrm{Hom}_P(Q, X)$, so they agree after composing with $X_k \to X_{k_1}$ for some $k_1 \geq k$. View the $X_{k'}$, $k' \geq k_1$, under Q via this common restriction; the images of c_1, c_2 in X_{k_1} are maps under Q with the same image in $\mathrm{Hom}_Q(Q', X)$, so they agree in some X_{k_2}, $k_2 \geq k_1$.

(c) By Lemma 4.5(b), $\mathrm{Hom}_P(P[-x]^\sharp, X)$ is a point if $x \mapsto 0$ in X and empty otherwise; in a filtered colimit, $x \mapsto 0$ holds iff it holds at some stage (Lemma 4.2). □

Lemma 4.7 (Retract of a finite presentation)**.** *Let $B \in \mathcal{C}$ and let $B \to M$ be f.p. Then there are a finite set S and a finite set R of relations in $B \oplus \mathbb{N}^S$ such that M is a retract under B of $T := ((B \oplus \mathbb{N}^S)/R)^{\mathrm{int,sat,sharp}}$. (Here T is this object itself, not its image in M.)*

Proof. Consider pairs (S, R) with $S \subset M$ finite and R a finite set of pairs (a, b) of elements of $B \oplus \mathbb{N}^S$ having the same image in M under the obvious map. These pairs are directed by inclusion. Let $T(S, R)$ be the reflection (Construction 4.3); the maps $T(S, R) \to M$ and the transition maps exist by its universal property. Let $L = \operatorname{colim} T(S, R)$ (Lemma 4.2). $L \to M$ is surjective: $m \in M$ is the image of the generator of $T(\{m\}, \emptyset)$. $L \to M$ is injective: let x, y lie in a common $T(S, R)$ and have the same image in M. By construction of the reflection there is $n \geq 1$ with $nx = [a]$ and $ny = [b]$ for some $a, b \in B \oplus \mathbb{N}^S$. Then a, b have equal images in M, and in $T(S, R \cup \{(a, b)\})$ we get $nx = ny$, hence $x = y$ by Lemma 4.1. So $M = \operatorname{colim} T(S, R)$ in $B/\mathcal{C}$. Since M is f.p., id_M factors through some $T = T(S, R)$: there is $s\colon M \to T$ under B whose composite with $T \to M$ is id_M. □

Remark 4.8. Conversely, any T as in Lemma 4.7 is f.p. under B: by Construction 4.3, $\mathrm{Hom}_B(T, X)$ is the set of $y \in X^S$ satisfying the finitely many relations R, and this commutes with filtered colimits by Lemma 4.2.

5 The site

Definition 5.1. For an object Q of A_P, a *birhizomic cover of Q* is a family $\{a_i\colon Q \to Q_i\}$ of maps under P that is a birhizomic cover in the sense of the problem, with Q as base. Its *witness set* is $I_0 = \{i :$ there are a valuative V and $g\colon Q_i \to V$ such that a_i is birhizomic at $g\}$. By definition of "birhizomic at", a_i is f.p. for every $i \in I_0$.

Lemma 5.2 (Pretopology axioms)**.** *Birhizomic covers satisfy the following.*

(a) Isomorphisms are birhizomic covers.

(b) Base change: if $\{a_i\colon Q \to Q_i\}$ is a cover and $c\colon Q \to Q'$, then $\{Q' \to Q'_i := Q' \sqcup_Q Q_i\}$ is a cover.

(c) Composition: if $\{a_i\colon Q \to Q_i\}$ is a cover and each $\{b_{ij}\colon Q_i \to Q_{ij}\}$ is a cover, then $\{b_{ij}a_i\}$ is a cover.

Proof. (a) Let $\varphi\colon Q \to Q'$ be an isomorphism and v a valuation of Q. Then $v = (v\varphi^{-1})\varphi$ and φ is f.p. Given $h\colon Q \to M$ and an infinitesimal $p\colon M \to V$ with $ph = v$, the map $u = h\varphi^{-1}$ is the unique one with $u\varphi = h$, and $pu = v\varphi^{-1}$.

(b) Let v' be a valuation of Q'. Then $v'c = g_i a_i$ with a_i birhizomic at g_i. Put $g' = (v', g_i)\colon Q'_i \to V$. The map $Q' \to Q'_i$ is f.p. by Lemma 4.6(b). Let $h\colon Q' \to M$ and an infinitesimal $p\colon M \to V$ satisfy $ph = v'$. Then (hc, p) factors $g_i a_i$, so there is a unique $u_i\colon Q_i \to M$ with $pu_i = g_i$ and $u_i a_i = hc$. The map $u = (h, u_i)\colon Q'_i \to M$ has $pu = g'$ and restricts to h on Q'. Any such u restricts on Q_i to a map with the defining properties of u_i, which gives uniqueness.

(c) Let v be a valuation of Q. Write $v = g_i a_i$ with a_i birhizomic at g_i, and $g_i = g_{ij} b_{ij}$ with b_{ij} birhizomic at g_{ij}. The composite $b_{ij}a_i$ is f.p. by Lemma 4.6(a). Given (h, p) with p infinitesimal and $ph = v$, take the unique u_i with $pu_i = g_i$ and $u_i a_i = h$, then the unique u with $pu = g_{ij}$ and $ub_{ij} = u_i$; it satisfies $ub_{ij}a_i = h$. If u' has these properties, then $u'b_{ij}$ has those of u_i, so $u'b_{ij} = u_i$, and then $u' = u$. □

Hence (by the standard passage from a pretopology to a topology [1, Exp. II, §1]) the Grothendieck topology $\mathcal{J}$ that the birhizomic covers define on A_P^{op} has as covering sieves of Q exactly the sieves containing (the sieve generated by) some birhizomic cover of Q, and its sheaves are the presheaves satisfying the sheaf condition for birhizomic covers. The property we use is:

($\star$) *every covering sieve of every object Q contains all members of some birhizomic cover of Q.*

Lemma 5.3 (Witness subfamily)**.** *$\{Q \to Q_i\}_{i\in I_0}$ is a birhizomic cover of Q, all of whose members are f.p.*

Proof. For a valuation v of Q, the cover property gives i and a factorization through Q_i at which a_i is birhizomic; then $i \in I_0$. □

6 Kernel faces of finitely presented maps are principal

Theorem 6.1. *For every monoid P and every f.p. map $f\colon P \to Q$, there is $s \in P$ with $\Phi_Q = \mathrm{face}(s)$.*

Proof. Step 1: reduction. By Lemma 4.7 with $B = P$, Q is a retract under P of $Q' = ((P \oplus \mathbb{N}^k)/R)^{\mathrm{int,sat,sharp}}$ with R finite. Then $\ker(P \to Q) = \ker(P \to Q')$: if x dies in Q', it dies in Q, the image of Q' under the retraction; if x dies in Q, apply the section $Q \to Q'$, which is a map under P.

Step 2: the kernel of $P \to Q'$. Write the relations as pairs (r_j, s_j) in $P \oplus \mathbb{N}^k$, $j = 1, \dots, \ell$, and put $\lambda_j = r_j - s_j \in P^{\mathrm{gp}} \oplus \mathbb{Z}^k$. Let L be the subgroup they generate and $G = (P^{\mathrm{gp}} \oplus \mathbb{Z}^k)/L$. As in Construction 4.3, M_0 is the image of $P \oplus \mathbb{N}^k$ in G, M_1 is its saturation in G, and $Q' = M_1/M_1^*$. Now $x \in P$ dies in Q' iff its image is a unit of M_1, iff $-x \in M_1$, iff $-nx \in M_0$ for some $n \geq 1$, iff $(-nx - p, -m) \in L$ for some $p \in P$, $m \in \mathbb{N}^k$. Write the element of L as $\sum_j c_j\lambda_j$ with $c \in \mathbb{Z}^\ell$.

- The $\mathbb{Z}^k$-component condition says that c lies in $\Lambda := \{c \in \mathbb{Z}^\ell : \text{the } \mathbb{Z}^k\text{-part of } \sum_j c_j\lambda_j \text{ lies in } -\mathbb{N}^k\}$.
- Λ is the set of lattice points of a rational polyhedral cone, so it is a finitely generated monoid by Gordan's lemma [3], [2, §1.2].
- Let $\psi\colon \Lambda \to P^{\mathrm{gp}}$ be the P^{gp}-part of $c \mapsto -\sum_j c_j\lambda_j$. It is additive, so $Y := \psi(\Lambda)$ is a finitely generated submonoid of P^{gp}.
- The condition reads $nx + p = \psi(c)$. Conversely, for $c \in \Lambda$ with $\psi(c) = nx + p$, put $m := -(\mathbb{Z}^k\text{-part of } \sum_j c_j\lambda_j) \in \mathbb{N}^k$.

Hence x dies iff $nx + p \in Y$ for some $n \geq 1$ and $p \in P$, i.e. iff $nx \leq y$ for some $y \in S := Y \cap P$. Since $x \leq nx$, this holds iff $x \leq y$ for some $y \in S$. So Φ_Q is the face generated by the submonoid S, namely $\bigcup_{t\in S} \mathrm{face}(t)$.

Step 3: principality. The family $\{\mathrm{face}(t) : t \in S\}$ is directed, since $\mathrm{face}(t + t') \supset \mathrm{face}(t), \mathrm{face}(t')$. Consider a strictly increasing chain $\mathrm{face}(s_1) \subsetneq \mathrm{face}(s_2) \subsetneq \cdots$ with $s_i \in S$, and put $G_i := \mathrm{face}(s_i)^{\mathrm{gp}} \cap Y^{\mathrm{gp}}$.

- For any face F of P, $F^{\mathrm{gp}} \cap P = F$: if $x = a - b$ with $a, b \in F$ and $x \in P$, then $x + b = a \in F$, so $x \in F$.
- If $ns_{i+1} \in G_i$ for some $n \geq 1$, then $ns_{i+1} \in \mathrm{face}(s_i)^{\mathrm{gp}} \cap P = \mathrm{face}(s_i)$, so $s_{i+1} \in \mathrm{face}(s_i)$, contradicting strictness.

- So $s_{i+1} \in G_{i+1}$ has infinite order modulo G_i. Since $G_i \subset G_{i+1}$, the ranks strictly increase inside the finitely generated free abelian group Y^{gp} (P^{gp} is torsion-free).

Hence chains have length at most $\operatorname{rk} Y^{\mathrm{gp}} + 1$, and there is a maximal face(s) with $s \in S$ (start from $s = 0$). By directedness it contains every face(t), $t \in S$, so $\Phi_Q = \mathrm{face}(s)$. □

7 The nerve theorem

Definition 7.1 (Ordered complexes)**.** Let N be a simplicial complex on a vertex set I. Its *augmented ordered chain complex* $\tilde{C}(N)$ has $\tilde{C}_p$ free on the tuples $(i_0, \dots, i_p) \in I^{p+1}$ (repetitions allowed) with $\{i_0, \dots, i_p\} \in N$, for $p \geq 0$, and $\tilde{C}_{-1} = \mathbb{Z}$; the differential is $d = \sum_j (-1)^j (\text{delete } i_j)$, each vertex mapping to $1 \in \tilde{C}_{-1}$. The *augmented ordered cochain complex* with coefficients in an abelian group A is $\mathrm{Hom}(\tilde{C}(N), A)$. For a tuple t, $\mathrm{set}(t)$ denotes its set of entries.

Theorem 7.2. *Let P be any monoid, $f\colon P \to Q$ any map in $\mathcal{C}$, and $\{Q \to Q_i\}_{i \in I}$ a family of f.p. maps under Q such that every valuation $v\colon Q \to V$ (V valuative) factors through some Q_i. For finite nonempty $\sigma \subset I$ let Q_σ be the pushout of the Q_i ($i \in \sigma$) over Q in $\mathcal{C}$, and put $\Phi_\sigma := \ker(P \to Q_\sigma)$. Let $\tau \subset P$ be any subset with $\Phi_Q \subset \tau$, and put $N_\tau := \{\sigma : \Phi_\sigma \subset \tau\}$. Then:*

(a) N_τ is a nonempty simplicial complex (closed under nonempty subsets, since $Q_{\sigma'} \to Q_\sigma$ for $\sigma' \subset \sigma$);

(b) its augmented ordered chain complex is exact;

(c) its augmented ordered cochain complex with coefficients in any abelian group A is exact.

The proof occupies the rest of this section.

7.1 Cones and valuations

Put $U := Q^{\mathrm{gp}} \otimes \mathbb{Q}$; then $Q \subset U$ (Lemma 4.1). Let $Q_{\mathbb{Q}} := \mathbb{Q}_{\geq 0} \cdot Q$. A *cone* is a subset $K \subset U$ with $0 \in K$, $K + K \subset K$ and $\mathbb{Q}_{>0} K \subset K$; its *lineality* is the subspace $\operatorname{lin} K := K \cap (-K)$. A cone W is *total* if $W \cup (-W) = U$. For $X \subset U$, $\mathrm{cone}(X)$ is the set of finite $\mathbb{Q}_{\geq 0}$-combinations of elements of X ($\mathrm{cone}(\emptyset) = \{0\}$).

- $\operatorname{lin} Q_{\mathbb{Q}} = 0$: if $x = q/n$ and $-x = q'/n'$ with $q, q' \in Q$, then $n'q + nq' = 0$ in Q^{gp}, so $q = q' = 0$ by sharpness.
- A valuation $v\colon Q \to V$ gives the total cone $W_v := \{x \in U : v(x) \geq 0 \text{ in } V^{\mathrm{gp}} \otimes \mathbb{Q}\} \supset Q_{\mathbb{Q}}$. Here v is extended $\mathbb{Q}$-linearly, and the divisible hull $V^{\mathrm{gp}} \otimes \mathbb{Q}$ of the totally ordered group V^{gp} is totally ordered.
- Conversely, a total cone $W \supset Q_{\mathbb{Q}}$ gives a valuation $v_W\colon Q \to V_W := W/\operatorname{lin} W$. Here V_W is the nonnegative part of the totally ordered $\mathbb{Q}$-vector space $H_W := U/\operatorname{lin} W$, so it is integral, saturated, sharp and valuative. Moreover $v_W(x) \geq 0$ iff $x \in W$, and $v_W(x) = 0$ iff $x \in \operatorname{lin} W$ (for $x \in U$).

Lemma 7.3. *Every cone K is contained in a total cone W with $\operatorname{lin} W = \operatorname{lin} K$.*

Proof. Cones $K' \supset K$ with $\operatorname{lin} K' = \operatorname{lin} K$ are closed under unions of chains, so Zorn's lemma gives a maximal one, K'. Suppose $x \notin K' \cup (-K')$ and put $K'' = K' + \mathbb{Q}_{\geq 0} x$. If $y \in \operatorname{lin} K''$, write $y = k_1 + ax$ and $-y = k_2 + bx$ with $k_i \in K'$ and $a, b \geq 0$. Then $k_1 + k_2 = -(a+b)x$. If $a + b > 0$, dividing gives $-x \in K'$, a contradiction. So $a = b = 0$ and $y \in \operatorname{lin} K'$. Thus $\operatorname{lin} K'' = \operatorname{lin} K'$ while $K'' \supsetneq K'$, contradicting maximality. So K' is total. □

Injective valuations. Every $M \in \mathcal{C}$ has a valuation $\iota\colon M \to V$ with $\iota(m) = 0$ only for $m = 0$: apply Lemma 7.3 to $M_{\mathbb{Q}} \subset M^{\mathrm{gp}} \otimes \mathbb{Q}$, which has lineality 0, and take v_W.

7.2 Lifting loci are given by finitely many sign conditions

For $i \in I$ let E_i be the set of total cones $W \supset Q_{\mathbb{Q}}$ such that v_W extends to Q_i, i.e. there is $g\colon Q_i \to V_W$ under Q.

Lemma 7.4. *For each i there is a finite set $X_i \subset U$ such that:*

(a) $E_i = \{W : X_i \subset W\}$;

(b) for every valuation $v\colon Q \to V$, if v extends to Q_i then $X_i \subset W_v$.

Proof. (i) By Lemma 4.7 with $B = Q$, $M := Q_i$ is a retract under Q of $T = ((Q \oplus \mathbb{N}^S)/R)^{\mathrm{int,sat,sharp}}$ with S, R finite. A map out of Q extends (under Q) to M iff it extends to T: compose with $T \to M$ in one direction and with $M \to T$ in the other.

(ii) Let H be a totally ordered $\mathbb{Q}$-vector space and $v\colon Q \to H_{\geq 0}$ a map. Since $H_{\geq 0} \in \mathcal{C}$, by Construction 4.3 a map $T \to H_{\geq 0}$ under Q is the same as a vector $y \in H^S$ with $y \geq 0$ satisfying the relations. Write the j-th relation as $(a_j, \alpha_j) = (b_j, \beta_j)$ with $a_j, b_j \in Q$ and $\alpha_j, \beta_j \in \mathbb{N}^S$. It reads $(\alpha_j - \beta_j) \cdot y = c_j$, where $c_j := v(b_j - a_j)$ and v is extended $\mathbb{Q}$-linearly to U.

(iii) *Fourier–Motzkin elimination* (cf. [7]). The system $\{y \geq 0;\ (\alpha_j - \beta_j) \cdot y - c_j = 0\}$ is a finite homogeneous system of non-strict linear inequalities with rational coefficients in the unknowns (y, c), each equation counting as two inequalities. Eliminating one unknown y_s replaces the system by all sums of a lower bound and an upper bound for y_s (after scaling by positive rationals), together with the inequalities not involving y_s. This preserves solvability over any totally ordered $\mathbb{Q}$-vector space H: given the other unknowns, take y_s to be the maximum of its finitely many lower bounds, or the minimum of its upper bounds if there are no lower bounds, or 0 if there are neither. Eliminating all of y gives finitely many rational vectors u_m, obtained by a formal procedure independent of H, such that

$$\text{the system is solvable in } H \iff u_m \cdot c \geq 0 \text{ for all } m.$$

Put $x_m := \sum_j u_{mj}(b_j - a_j) \in U$ and $X_i := \{x_m\}$. Then $u_m \cdot c = v(x_m)$.

(a) Take $H = H_W$ and $v = v_W$. Then v_W extends to Q_i iff $v_W(x_m) \geq 0$ for all m, iff $X_i \subset W$.

(b) Take $H = V^{\mathrm{gp}} \otimes \mathbb{Q}$, which contains V in $H_{\geq 0}$. If v extends to Q_i, it extends to T with values in $V \subset H_{\geq 0}$, so $v(x_m) \geq 0$ for all m, i.e. $X_i \subset W_v$. □

7.3 The nerve condition as a cone condition

For finite σ put $X_\sigma := \bigcup_{i \in \sigma} X_i$ and $K_\sigma := Q_{\mathbb{Q}} + \mathrm{cone}(X_\sigma)$. For a cone $K \supset Q_{\mathbb{Q}}$ put $f^{-1}(\mathrm{lin}\, K) := \{p \in P : f(p) \in \mathrm{lin}\, K\}$; since $f(p) \in Q_{\mathbb{Q}} \subset K$, this is $\{p : -f(p) \in K\}$.

Lemma 7.5. *$\Phi_\sigma = f^{-1}(\mathrm{lin}\, K_\sigma)$. The same holds for an ordered tuple $t = (i_0, \ldots, i_p)$ with repetitions and $Q_t := Q_{i_0} \sqcup_Q \cdots \sqcup_Q Q_{i_p}$: $\Phi_t := \ker(P \to Q_t)$ equals $\Phi_{\mathrm{set}(t)}$.*

Proof. A map $Q_\sigma \to V_W$ under Q is a family of maps $Q_i \to V_W$ under Q, $i \in \sigma$, because Q_σ is the pushout in $\mathcal{C}$ and $V_W \in \mathcal{C}$. So v_W extends to Q_σ iff $W \in \bigcap_{i \in \sigma} E_i$, iff $K_\sigma \subset W$ (Lemma 7.4(a)).

($\subset$) Let $p \in \Phi_\sigma$. Take $W \supset K_\sigma$ total with $\mathrm{lin}\, W = \mathrm{lin}\, K_\sigma$ (Lemma 7.3). Then v_W extends to Q_σ and kills the image of p, so $f(p) \in \mathrm{lin}\, W = \mathrm{lin}\, K_\sigma$.

($\supset$) Let $f(p) \in \operatorname{lin} K_\sigma$. Take an injective valuation $\iota\colon Q_\sigma \to V$ and let v be its restriction to Q. Then v extends to each Q_i, $i \in \sigma$, so by Lemma 7.4(b), $X_\sigma \subset W_v$ and $K_\sigma \subset W_v$. Hence $v(f(p)) = 0$ in $V^{\mathrm{gp}} \otimes \mathbb{Q}$, so it is 0 in V (Lemma 4.1). Thus ι kills the image $\bar{p}$ of p in Q_σ, and $\bar{p} = 0$.

For tuples, maps out of Q_t are again tuples of maps out of the Q_{i_j}, so the same argument gives $\Phi_t = f^{-1}(\operatorname{lin} K_{\mathrm{set}(t)}) = \Phi_{\mathrm{set}(t)}$. □

Let $\mathcal{G} := \{\text{cones } K \supset Q_\mathbb{Q} : f^{-1}(\operatorname{lin} K) \subset \tau\}$. By Lemma 7.5, $N_\tau = \{\sigma : K_\sigma \in \mathcal{G}\}$, and:

(G0) if $Q_\mathbb{Q} \subset K \subset K'$ and $K' \in \mathcal{G}$, then $K \in \mathcal{G}$;

(G1) if $K \in \mathcal{G}$ and $W \supset K$ is a cone with $\operatorname{lin} W = \operatorname{lin} K$, then $W \in \mathcal{G}$;

(G2) for a cone $K \supset Q_\mathbb{Q}$ and $z \in U$, $f^{-1}(\operatorname{lin}(K + \mathbb{Q}z)) = f^{-1}(\operatorname{lin}(K + \mathbb{Q}_{\geq 0}z)) \cup f^{-1}(\operatorname{lin}(K + \mathbb{Q}_{\geq 0}(-z)))$ (write $-f(p) = k + tz$ and split on the sign of t); hence if $K + \mathbb{Q}_{\geq 0}z$ and $K + \mathbb{Q}_{\geq 0}(-z)$ are both in $\mathcal{G}$, so is $K + \mathbb{Q}z$;

(G3) $Q_\mathbb{Q} \in \mathcal{G}$, because $f^{-1}(\operatorname{lin} Q_\mathbb{Q}) = f^{-1}(0) = \Phi_Q \subset \tau$.

7.4 A finite subcover

Lemma 7.6. *There is a finite $J_0 \subset I$ such that every total cone $W \supset Q_\mathbb{Q}$ lies in $\bigcup_{i \in J_0} E_i$.*

Proof. The set $\mathcal{T}$ of total cones $W \supset Q_\mathbb{Q}$ is a closed subset of $\{0,1\}^U$, since each defining condition involves finitely many coordinates: $z \in W$ or $-z \in W$; $z, z' \in W \Rightarrow z + z' \in W$; $z \in W \Rightarrow rz \in W$ ($r \in \mathbb{Q}_{>0}$); $q \in W$ for $q \in Q_\mathbb{Q}$. So $\mathcal{T}$ is compact by Tychonoff's theorem. Each $E_i = \{W : X_i \subset W\}$ is open since X_i is finite. The E_i cover $\mathcal{T}$: v_W is a valuation, so it factors through some Q_i, i.e. $W \in E_i$. Take a finite subcover. □

7.5 Abstract acyclicity

Fix a $\mathbb{Q}$-vector space U, a cone $K_0 \subset U$, a family $\mathcal{G}$ of cones containing K_0 satisfying (G0), (G1) and (G2) for cones containing K_0, a finite family $(X_j)_{j \in J}$ of finite subsets of U, and an abelian group A. For a tuple $\sigma = (j_0, \ldots, j_p)$ write $\operatorname{cone} X_\sigma := \operatorname{cone} X_{j_0} + \cdots + \operatorname{cone} X_{j_p}$ (and $\{0\}$ for the empty tuple). For a cone K' let $A(K') = A$ if $K' \in \mathcal{G}$ and $A(K') = 0$ otherwise. For a cone $K \supset K_0$ let $C^\bullet(K)$ be the complex with

$$C^p(K) = \prod_{\sigma \in J^{p+1}} A(K + \operatorname{cone} X_\sigma) \qquad (p \geq -1),$$

where the term for $p = -1$ is $A(K)$, the component maps are id_A when both ends are A and 0 otherwise (this is well defined by (G0), since a coface map goes to a larger cone, so a pattern $0 \to A$ never occurs), and d is the alternating sum of the deletions. If $K \in \mathcal{G}$, $C^\bullet(K)$ is the augmented ordered cochain complex of the simplicial complex $N(K) := \{\sigma : K + \operatorname{cone} X_\sigma \in \mathcal{G}\}$. If $K \notin \mathcal{G}$, $C^\bullet(K) = 0$ by (G0).

Theorem 7.7. *If every total cone $W \supset K$ contains some X_j, then $C^\bullet(K)$ is exact.*

Proof. Let $Z := \bigcup_j X_j$ (finite); call $z \in Z$ *undecided* for K if $z \notin K$ and $-z \notin K$. We induct on the number $m(K)$ of undecided elements. We may assume $K \in \mathcal{G}$.

Cone lemma. Suppose there is j^* with $\{j^*\} \in N(K)$ such that $\sigma \in N(K) \Rightarrow \sigma \cup \{j^*\} \in N(K)$. Define $(hc)(j_0, \ldots, j_{p-1}) := c(j^*, j_0, \ldots, j_{p-1})$. Then h is well defined: if the target tuple carries A, its set lies in $N(K)$, so the set with j^* added does too, and the source tuple carries A; if the target

carries 0 there is nothing to check. The standard computation gives $dh + hd = \mathrm{id}$: on a tuple t carrying A, every tuple involved has its set inside $\mathrm{set}(t) \cup \{j^*\} \in N(K)$, so all component maps are identities; in degree -1, $(hdc)() = (dc)(j^*) = c()$, which uses $\{j^*\} \in N(K)$ and $K \in \mathcal{G}$. So $C^\bullet(K)$ is contractible.

Base case $m(K) = 0$. Take $W \supset K$ total with $\lin W = \lin K$ (Lemma 7.3) and some $X_{j^*} \subset W$. Let $z \in X_{j^*}$. If $z \notin K$, then $-z \in K$ because z is decided, so $z \in \lin W = \lin K \subset K$, a contradiction. So $X_{j^*} \subset K$ and $K + \operatorname{cone} X_{j^*} = K$. Hence j^* is a cone point and the cone lemma applies.

Row lemma. For any cone $K' \supset K_0$ and $z \in U$, the complex built as above from K' and the two-member family $X_+ = \{z\}$, $X_- = \{-z\}$ is exact. If $K' \notin \mathcal{G}$ it is zero. Otherwise some total $W \supset K'$ with $\lin W = \lin K'$ lies in $\mathcal{G}$ (Lemma 7.3 and (G1)); W contains z or $-z$, so $K' + \mathbb{Q}_{\geq 0} z$ or $K' + \mathbb{Q}_{\geq 0}(-z)$ lies in $\mathcal{G}$ by (G0), i.e. one vertex is in the nerve; and if both vertices are, so is the edge, because $K' + \operatorname{cone}\{z, -z\} = K' + \mathbb{Q} z$ (G2). So the nerve is a vertex or an edge, and the cone lemma applies.

Inductive step, $m(K) > 0$. Pick an undecided $z \in Z$ and form the double complex

$$D^{p,q} = \prod_{\sigma \in J^{p+1},\ \rho \in \{\pm\}^{q+1}} A\big(K + \operatorname{cone} X_\sigma + \operatorname{cone}\{\rho_0 z, \dots, \rho_q z\}\big), \qquad p, q \geq -1,$$

with the two commuting deletion differentials.

- *Rows.* For fixed p, the row is the product over σ of row-lemma complexes on $K + \operatorname{cone} X_\sigma$, hence exact.
- *Columns* $q \geq 0$. The column is $\prod_\rho C^\bullet(K_\rho)$ with $K_\rho := K + \operatorname{cone}\{\rho_0 z, \dots, \rho_q z\} \in \{K + \mathbb{Q}_{\geq 0} z,\ K + \mathbb{Q}_{\geq 0}(-z),\ K + \mathbb{Q} z\}$. The cover hypothesis passes to $K_\rho \supset K$, and $m(K_\rho) < m(K)$ because z becomes decided and decided elements stay decided. So these columns are exact by induction.
- *Column* $q = -1$ is $C^\bullet(K)$.

Each total degree involves finitely many (p, q). Filtering by p, with exact rows, shows that $\operatorname{Tot} D$ is acyclic (acyclic assembly lemma [8, §2.7]). The columns $q \geq 0$ form a subcomplex of $\operatorname{Tot} D$, since the vertical differential raises q; filtering it by q shows it is acyclic. The quotient is column -1, i.e. $C^\bullet(K)$ up to shift and sign, so the long exact sequence shows that $C^\bullet(K)$ is exact. □

7.6 Proof of Theorem 7.2

1. *Finite pieces.* Let $J' \supset J_0$ be finite. Apply Theorem 7.7 with $U = Q^{\mathrm{gp}} \otimes \mathbb{Q}$, $K_0 = K = Q_\mathbb{Q}$, the family $\mathcal{G}$ above, and $(X_i)_{i \in J'}$. Every total $W \supset Q_\mathbb{Q}$ contains some X_i with $i \in J_0$, by Lemmas 7.6 and 7.4(a). By Lemma 7.5 and (G3), $N(Q_\mathbb{Q})$ is the full subcomplex $N_\tau(J')$ of N_τ on J', and $C^\bullet(Q_\mathbb{Q})$ is its augmented ordered cochain complex. So this complex is exact for every A.

2. *Nonemptiness.* Take $W_0 \supset Q_\mathbb{Q}$ total with $\lin W_0 = 0$ (Lemma 7.3). Then $W_0 \in \mathcal{G}$ by (G3) and (G1). W_0 contains some X_i, so $K_{\{i\}} \subset W_0$ and $\{i\} \in N_\tau$ by (G0).

3. *Chains, finite case.* $\tilde{C} := \tilde{C}(N_\tau(J'))$ is a complex of free abelian groups. The universal coefficient theorem [5, §3.1] gives a surjection $H^n(\mathrm{Hom}(\tilde{C}, A)) \to \mathrm{Hom}(H_n \tilde{C}, A)$. Taking $A = H_n \tilde{C}$ shows $H_n \tilde{C} = 0$.

4. *Chains, general case.* $\tilde{C}(N_\tau)$ is the directed union of the $\tilde{C}(N_\tau(J'))$ over finite $J' \supset J_0$, so it is exact.

5. *Cochains, general case.* An exact complex of free abelian groups is split: each boundary group $B_{n-1} \subset \tilde{C}_{n-1}$ is free, so $0 \to Z_n \to \tilde{C}_n \to B_{n-1} \to 0$ splits. Hence $\mathrm{Hom}(\tilde{C}(N_\tau), A)$ is exact for every A. □

8 The sheaf-theoretic part

Presheaves on A_P^{op} (resp. B_P^{op}) are functors $A_P \to \mathrm{Ab}$ (resp. $B_P \to \mathrm{Ab}$); $\iota\colon B_P \to A_P$ is the inclusion. Since the topology on B_P^{op} is trivial, exactness in $\mathrm{Sh}(B_P) = \mathrm{PSh}(B_P)$ is objectwise.

8.1 Kan extension, π^{-1} and π_*

The left Kan extension is $(\mathrm{Lan}\, F)(Q) = \mathrm{colim}_{(b,\, b\to Q)\in \iota/Q} F(b)$ [6, Ch. X]. By Lemma 4.5, a pair $(b_x, b_x \to Q)$ exists iff $x \in \Phi_Q$, and the map is then unique. So ι/Q is equivalent to the poset $\{\mathrm{face}(x) : x \in \Phi_Q\}$, which is nonempty (it contains $\{0\}$) and directed ($x, y \in \Phi_Q \Rightarrow x + y \in \Phi_Q$). Hence $(\mathrm{Lan}\, F)(Q)$ is a filtered colimit, and Lan **is exact.**

Lemma 8.1. *Let Q be f.p. under P. By Theorem 6.1, $\Phi_Q = \mathrm{face}(x_Q)$ for some x_Q; put $b(Q) := b_{x_Q}$.*

(a) $(b(Q), b(Q) \to Q)$ is terminal in ι/Q, so $F(\Phi_Q) := F(b(Q)) \to (\mathrm{Lan}\, F)(Q)$ is an isomorphism, natural in F.

(b) For a map $Q \to Q'$ of f.p. objects, $(\mathrm{Lan}\, F)(Q \to Q')$ corresponds to $F(b(Q) \to b(Q'))$.

(c) For $b \in B_P$, the unit $F(b) \to (\mathrm{Lan}\, F)(b)$ is an isomorphism.

(d) $(\mathrm{Lan} \prod_j F_j)(Q) = \prod_j (\mathrm{Lan}\, F_j)(Q)$ naturally for f.p. Q.

Proof. (a) $b(Q) \to Q$ exists since $x_Q \in \Phi_Q$. If $b_x \to Q$, then $x \in \Phi_Q = \mathrm{face}(x_Q)$, so there is a unique map $b_x \to b(Q)$ (Lemma 4.5(e)), automatically compatible with the maps to Q by uniqueness. (b) $\Phi_Q \subset \Phi_{Q'}$ gives the map $b(Q) \to b(Q')$, compatible with the colimit maps. (c) b is terminal in ι/b. (d) Both sides are evaluation at $b(Q)$ by (a). □

π^{-1} and π_*. $\pi^{-1}F = (\mathrm{Lan}\, F)^{\#}$, where $\#$ is sheafification for $\mathcal{J}$; it is exact since Lan and $\#$ are. For a sheaf G, $\mathrm{Hom}(\pi^{-1}F, G) = \mathrm{Hom}(\mathrm{Lan}\, F, G) = \mathrm{Hom}(F, G \circ \iota)$, so $\pi_* G = G|_{B_P}$. The unit $F \to \pi_*\pi^{-1}F$ at b is the composite $F(b) \cong (\mathrm{Lan}\, F)(b) \to (\mathrm{Lan}\, F)^{\#}(b)$.

8.2 Injectives

For $\tau \in \mathrm{Pr}(P)$ and an abelian group A, let $I_{\tau,A}(b) = A$ if $\Phi_b \subset \tau$ and 0 otherwise, with transition maps id_A or 0. This is a functor on B_P: $b \to b'$ gives $\Phi_b \subset \Phi_{b'}$, so a pattern $0 \to A$ never occurs.

Lemma 8.2. $\mathrm{Hom}(F, I_{\tau,A}) \cong \mathrm{Hom}(F(\tau), A)$. *Hence $I_{\tau,A}$ is injective when A is.*

Proof. If $\Phi_b \subset \tau$, there is a unique map $b \to b_\tau$, and naturality forces $\varphi_b = \varphi_{b_\tau} \circ F(b \to b_\tau)$. Conversely this formula, with $\varphi_b = 0$ otherwise, defines a natural map, because B_P is a preorder. So $\mathrm{Hom}(-, I_{\tau,A}) = \mathrm{Hom}(\mathrm{ev}_{b_\tau}(-), A)$, which is exact when A is injective. □

Resolution. Let $E(-)$ denote an injective embedding of abelian groups. Then $F \to \prod_{\tau\in \mathrm{Pr}(P)} I_{\tau, E(F(\tau))}$ is a monomorphism: at b, the component $\tau = \Phi_b$ is $F(b) \hookrightarrow E(F(b))$. Iterating on cokernels gives an objectwise exact injective resolution $0 \to F \to I^0 \to I^1 \to \cdots$ with $I^k = \prod_\tau I_{\tau, A_{k,\tau}}$ (possibly infinite products, $A_{k,\tau}$ injective).

8.3 The system Cov

Let $\mathfrak{B}$ be the set of f.p. objects of A_P, and let Cov be the set of birhizomic covers $\{Q \to Q_i\}$ with $Q \in \mathfrak{B}$ and every $Q \to Q_i$ f.p.

Lemma 8.3. *The system* Cov *has the following properties.*

(i) Every member of Cov *is a covering family for $\mathcal{J}$.*

(ii) For $\mathcal{U} \in$ Cov, every Q_i and every iterated pushout $Q_t = Q_{i_0} \sqcup_Q \cdots \sqcup_Q Q_{i_p}$ lies in $\mathfrak{B}$.

(iii) For $Q \in \mathfrak{B}$ and every covering sieve R of Q, there is $\mathcal{U} \in$ Cov all of whose members lie in R.

(iv) For $\mathcal{U} \in$ Cov, every valuation of Q factors through some member.

Moreover every $b \in B_P$ lies in $\mathfrak{B}$.

Proof. (i) and (iv) are clear; (ii) is Lemma 4.6. (iii) By $(\star)$, R contains a birhizomic cover, and its witness subfamily (Lemma 5.3) lies in Cov. The last claim is Lemma 4.6(c). □

8.4 Čech complexes

For a presheaf G and $\mathcal{U} = \{Q \to Q_i\}_{i \in I}$, the augmented Čech complex is $\check{C}^p(\mathcal{U}, G) = \prod_{t \in I^{p+1}} G(Q_t)$ for $p \geq 0$ and $\check{C}^{-1} = G(Q)$, with the alternating restriction differential (all ordered tuples, repetitions allowed). $\check{H}^p(\mathcal{U}, G)$ denotes the cohomology of the non-augmented complex ($p \geq 0$).

Proposition 8.4. *Let $Q \in \mathfrak{B}$, $\mathcal{U} \in$ Cov over Q, $\tau \in \Pr(P)$ and $G = \operatorname{Lan} I_{\tau,A}$. If $\Phi_Q \not\subset \tau$, the augmented Čech complex of G on $\mathcal{U}$ is 0; otherwise it is the augmented ordered cochain complex of N_τ with coefficients A (notation of Theorem 7.2). In both cases it is exact.*

Proof. All Q_t are f.p. (Lemma 8.3(ii)), so $G(Q_t) = I_{\tau,A}(\Phi_{Q_t})$ by Lemma 8.1, which is A iff $\Phi_{Q_t} \subset \tau$; the restriction maps between nonzero terms are id_A. If $\Phi_Q \not\subset \tau$, every term vanishes since $\Phi_Q \subset \Phi_{Q_t}$. Otherwise $\Phi_{Q_t} = \Phi_{\mathrm{set}(t)}$ by Lemma 7.5, applied to the composite $P \to Q$ (and Q_t is the same pushout), so the complex is $\mathrm{Hom}(\tilde{C}(N_\tau), A)$. Exactness is Theorem 7.2, whose hypotheses hold: the members are f.p. under Q, every valuation of Q factors through one (Lemma 8.3(iv)), and $\Phi_Q \subset \tau$. □

Corollary 8.5. *For $Q \in \mathfrak{B}$ and $\mathcal{U} \in$ Cov over Q, the augmented Čech complex of each $\operatorname{Lan} I^k$ on $\mathcal{U}$ is exact.*

Proof. Only objects of $\mathfrak{B}$ occur, and on $\mathfrak{B}$, $\operatorname{Lan} I^k = \prod_\tau \operatorname{Lan} I_{\tau,A_{k,\tau}}$ naturally (Lemma 8.1(d)). Products of exact complexes are exact. □

8.5 Sheafification does not change values on f.p. objects

We use the standard facts [1, Exp. II, §3]: for a sieve R generated by a family $\{U_i \to U\}$ with pushouts, $\mathrm{Hom}_{\mathrm{PSh}}(R, G)$ is the equalizer of $\prod G(U_i) \rightrightarrows \prod G(U_{ij})$; $G^+(U) = \operatorname{colim}_{R \in \mathcal{J}(U)} \mathrm{Hom}(R, G)$; and $G^\# = G^{++}$.

Lemma 8.6. *Let G be a presheaf such that for all $Q \in \mathfrak{B}$ and $\mathcal{U} \in$ Cov over Q the sequence $0 \to G(Q) \to \prod G(Q_i) \rightrightarrows \prod G(Q_{ij})$ is exact (an equalizer). Then $G(Q) \to G^\#(Q)$ is an isomorphism for every $Q \in \mathfrak{B}$.*

Proof. $\mathcal{J}(Q)$ is directed under reverse inclusion (the intersection of two covering sieves is covering), and $G^+(Q)$ is the colimit over it. By Lemma 8.3(iii), the sieves generated by members of Cov are cofinal in $\mathcal{J}(Q)$; a cofinal subset of a directed set is directed and computes the same colimit. On each of them $\mathrm{Hom}(R, G) = G(Q)$, compatibly with the transition maps, so $G(Q) \cong G^+(Q)$. For $\mathcal{U} \in \mathrm{Cov}$, all of Q, Q_i, Q_{ij} lie in $\mathfrak{B}$, where $G \to G^+$ is an isomorphism, so G^+ satisfies the same hypothesis. Repeating gives the claim for $G^{++} = G^{\#}$. □

Corollary 8.7. *The following hold.*

(a) $\mathrm{Lan}\, F$ *satisfies the hypothesis of Lemma 8.6.*

(b) For $Q \in \mathfrak{B}$, $F(\Phi_Q) \cong (\mathrm{Lan}\, F)(Q) \cong (\pi^{-1}F)(Q)$, *naturally. In particular the unit* $F \to \pi_*\pi^{-1}F$ *is an isomorphism.*

(c) $(\pi^{-1}I^k)(Q) = I^k(\Phi_Q)$ *on* $\mathfrak{B}$, *and* $\check{H}^p(\mathcal{U}, \pi^{-1}I^k) = 0$ *for* $p > 0$ *and* $\mathcal{U} \in \mathrm{Cov}$.

Proof. (a) Use the exact sequence $0 \to \mathrm{Lan}\, F \to \mathrm{Lan}\, I^0 \to \mathrm{Lan}\, I^1$ and Corollary 8.5 in degrees $-1, 0, 1$. If $s \in (\mathrm{Lan}\, F)(Q)$ restricts to 0 on every Q_i, its image in $(\mathrm{Lan}\, I^0)(Q)$ does too, hence is 0, so $s = 0$. A compatible family (s_i) glues in $\mathrm{Lan}\, I^0$ to some t; the image of t in $(\mathrm{Lan}\, I^1)(Q)$ has all restrictions 0, hence is 0; so t comes from some $s \in (\mathrm{Lan}\, F)(Q)$, and s restricts to s_i by injectivity of $(\mathrm{Lan}\, F)(Q_i) \to (\mathrm{Lan}\, I^0)(Q_i)$. (b) follows from (a), Lemma 8.6 and Lemma 8.1(a); the unit at b is $F(b) \cong (\mathrm{Lan}\, F)(b) \to (\mathrm{Lan}\, F)^{\#}(b)$ and $b \in \mathfrak{B}$. (c) The first statement is (b) for $F = I^k$. Hence on $\mathcal{U} \in \mathrm{Cov}$ the Čech complex of $\pi^{-1}I^k$ is isomorphic to that of $\mathrm{Lan}\, I^k$, which is exact by Corollary 8.5. □

8.6 A Cartan-type criterion

The following is a version of a classical criterion of H. Cartan for comparing Čech and derived functor cohomology; we include a proof adapted to the system Cov.

Lemma 8.8. *For an injective abelian sheaf* $\mathcal{I}$ *on* $(A_P^{\mathrm{op}}, \mathcal{J})$ *and any family* $\mathcal{U} = \{U \to U_i\}_{i \in I}$ *in* A_P *(pushouts exist),* $\check{H}^p(\mathcal{U}, \mathcal{I}) = 0$ *for* $p > 0$.

Proof. For an object V let $\mathbb{Z}_V$ be the free abelian presheaf on $h_V = \mathrm{Hom}_{A_P}(V, -)$. Let $K_p = \bigoplus_{t \in I^{p+1}} \mathbb{Z}_{U_t}$ with the alternating face differential; by Yoneda, $\mathrm{Hom}_{\mathrm{PSh}}(K_\bullet, G) = \check{C}^\bullet(\mathcal{U}, G)$. Evaluate at W: $K_\bullet(W)$ splits as a sum over $\psi\colon U \to W$, and the summand for ψ is the ordered chain complex of the full simplex (with repetitions) on $T_\psi = \{(i, \varphi) : \varphi\colon U_i \to W \text{ compatible with } \psi\}$. Prepending a fixed element of T_ψ contracts it in positive degrees, and H_0 is $\mathbb{Z}$ if $T_\psi \neq \emptyset$ and 0 otherwise. So $K_\bullet$ resolves the presheaf $\mathbb{Z}_R$ of the sieve R generated by $\mathcal{U}$. Sheafification is exact and $\mathrm{Hom}_{\mathrm{Sh}}(-, \mathcal{I})$ is exact, so $\check{C}^\bullet(\mathcal{U}, \mathcal{I}) = \mathrm{Hom}_{\mathrm{Sh}}(K_\bullet^{\#}, \mathcal{I})$ is exact in positive degrees. □

Proposition 8.9. *Let* G *be an abelian sheaf on* $(A_P^{\mathrm{op}}, \mathcal{J})$ *with* $\check{H}^p(\mathcal{U}, G) = 0$ *for all* $p > 0$ *and all* $\mathcal{U} \in \mathrm{Cov}$. *Then* $H^p(U, G) = 0$ *for all* $p > 0$ *and all* $U \in \mathfrak{B}$.

Proof. Embed $G \hookrightarrow \mathcal{I}$ with $\mathcal{I}$ an injective sheaf (abelian sheaves on a site form an abelian category with enough injectives [4]), and let $\mathcal{Q} = \mathcal{I}/G$ (sheaf quotient). Then $G(W) = \ker(\mathcal{I}(W) \to \mathcal{Q}(W))$ for all W.

(a) $\mathcal{I}(V) \to \mathcal{Q}(V)$ *is onto for* $V \in \mathfrak{B}$. Take $s \in \mathcal{Q}(V)$. It lifts locally on some covering sieve R. By Lemma 8.3(iii) choose $\mathcal{V} = \{V \to V_i\} \in \mathrm{Cov}$ with members in R and lifts t_i of $s|_{V_i}$. Then $c_{ij} := t_j - t_i$ on V_{ij} lies in $G(V_{ij})$ and is a Čech 1-cocycle for G. Since $\check{H}^1(\mathcal{V}, G) = 0$, $c = dg$ with $g \in \check{C}^0(\mathcal{V}, G)$. The family $(t_i - g_i)$ is compatible, so it glues in the sheaf $\mathcal{I}$ to some t, and t maps to s because $\mathcal{Q}$ is a sheaf and t, s have the same restrictions to all V_i.

(b) $\mathcal{Q}$ *satisfies the hypothesis.* For $\mathcal{U} \in \mathrm{Cov}$ all the U_t lie in $\mathfrak{B}$, so $0 \to \check{C}(\mathcal{U}, G) \to \check{C}(\mathcal{U}, \mathcal{I}) \to \check{C}(\mathcal{U}, \mathcal{Q}) \to 0$ is exact by (a). In the long exact sequence $\check{H}^p(\mathcal{U}, \mathcal{I}) \to \check{H}^p(\mathcal{U}, \mathcal{Q}) \to \check{H}^{p+1}(\mathcal{U}, G)$ the left term is 0 for $p \geq 1$ by Lemma 8.8 and the right term is 0 by hypothesis, so $\check{H}^p(\mathcal{U}, \mathcal{Q}) = 0$ for $p \geq 1$.

(c) *Conclusion.* From $0 \to G \to \mathcal{I} \to \mathcal{Q} \to 0$ and $H^{>0}(U, \mathcal{I}) = 0$ we get $H^1(U, G) = \mathrm{coker}(\mathcal{I}(U) \to \mathcal{Q}(U)) = 0$ by (a), and $H^p(U, G) \cong H^{p-1}(U, \mathcal{Q})$ for $p \geq 2$. Induction on p, applied to all sheaves satisfying the hypothesis at once, finishes the proof. □

8.7 Proof of the Main Theorem

By Corollary 8.7(c) and Proposition 8.9, $H^p(U, \pi^{-1} I^k) = 0$ for $p > 0$ and $U \in \mathfrak{B}$. Since π^{-1} is exact, $\pi^{-1}F \to \pi^{-1} I^\bullet$ is a resolution by $\Gamma(U, -)$-acyclic sheaves. So for $U \in \mathfrak{B}$, $H^p(U, \pi^{-1}F) = H^p(I^\bullet(\Phi_U))$ (Corollary 8.7(c)), which is $F(\Phi_U)$ for $p = 0$ and 0 for $p > 0$ because $I^\bullet$ is an objectwise exact resolution of F.

Since π_* is restriction, taking an injective resolution $G \to \mathcal{E}^\bullet$ of a sheaf G gives $(\mathrm{R}^p \pi_* G)(b) = H^p(\mathcal{E}^\bullet(b)) = H^p(b, G)$, as cohomology of presheaves on B_P is computed objectwise. Every $b \in B_P$ is f.p. (Lemma 8.3), hence $\mathrm{R}^p \pi_* \pi^{-1} F = 0$ for $p > 0$. On H^0, the derived unit $F \to \pi_* \pi^{-1} F \to \mathrm{R}\pi_* \pi^{-1} F$ induces the underived unit, which is an isomorphism by Corollary 8.7(b). So $F \to \mathrm{R}\pi_* \pi^{-1} F$ is a quasi-isomorphism. □

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